

Lab: Writing and Using a Python Function

Lab Objectives

Students will learn to:

- Write a Python function that accepts two numbers
- Display the output inside the function
- Add a `return` statement so the sum is returned to the caller
- Use **named arguments** so one or both values can be omitted
- Understand default parameters and named argument behavior

Exercise 1 — Write a basic `add_two` function

Write a function named `add_two` that:

1. Accepts **two numbers**
2. Computes their sum
3. Prints the sum
4. Does **not** return anything yet

Expected student code (no return yet):

```
def add_two(a, b):  
    result = a + b  
    print("The sum is:", result)
```

Call the function:

```
add_two(5, 7)
```

Exercise 2 — Add a return statement

Modify the function so that it:

- Still prints the sum
- **Also returns** the sum to the caller

Expected student code:

```
def add_two(a, b):
    result = a + b
    print("The sum is:", result)
    return result
```

Call the function and store the result:

```
value = add_two(10, 20)
print("Returned value:", value)
```

Exercise 3 — Add default values so arguments can be omitted

Modify the function so that:

- `a` defaults to 0
- `b` defaults to 0
- Students can call the function with **one**, **both**, or **no** arguments
- Named arguments can be used to specify only one value

Expected student code:

```
def add_two(a=0, b=0):
    result = a + b
    print("The sum is:", result)
    return result
```

Example calls:

<code>add_two()</code>	<code># uses both defaults → 0 + 0</code>
<code>add_two(a=5)</code>	<code># b defaults to 0 → 5 + 0</code>
<code>add_two(b=10)</code>	<code># a defaults to 0 → 0 + 10</code>
<code>add_two(a=3, b=4)</code>	<code># both provided → 3 + 4</code>

Solutions Section (Instructor Version)

✓ Solution for Exercise 1

```
def add_two(a, b):  
    result = a + b  
    print("The sum is:", result)  
  
add_two(5, 7)
```

✓ Solution for Exercise 2

```
def add_two(a, b):  
    result = a + b  
    print("The sum is:", result)  
    return result  
  
value = add_two(10, 20)  
print("Returned value:", value)
```

✓ Solution for Exercise 3 (Named Arguments + Defaults)

```
def add_two(a=0, b=0):  
    result = a + b  
    print("The sum is:", result)  
    return result  
  
add_two()           # 0 + 0  
add_two(a=5)        # 5 + 0  
add_two(b=10)        # 0 + 10  
add_two(a=3, b=4)    # 3 + 4
```